Pricing and Risk of Structured Products: Greeks, Monte Carlo and Hedging

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1 The Core Idea

A structured product is a bond plus options, and its risk is the risk of those options. A Phoenix autocall, for example, pays coupons, can be redeemed early, and loses capital if the underlying ends low. It has no closed-form price. A risk manager still needs its price, its sensitivities, and the effect of hedging it.

The repository (`structrisk`) does four things for this.

- **Closed-form Greeks** of a vanilla option, from first order (delta, vega) to third order (speed, zomma, color).
- **Monte Carlo pricing** of a Phoenix autocall and of other notes, with confidence intervals. Sensitivities come from bump-and-revalue with common random numbers.
- **Portfolio risk:** Greeks added across positions, in euros, with a spot-by-volatility stress grid.
- **Hedging simulation:** the replication error of a delta-hedged option, with and without costs.

2 Method

2.1 Black-Scholes-Merton Greeks

With dividend yield q, $d_1 = \dfrac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$, the call Greeks are:

Greek	Formula	Note
Delta	$e^{-qT}N(d_1)$	per unit of spot
Gamma	$\dfrac{e^{-qT}\varphi(d_1)}{S\sigma\sqrt{T}}$	same for call and put
Vega	$Se^{-qT}\varphi(d_1)\sqrt{T}$	per 1.00 of vol
Vanna	$-e^{-qT}\varphi(d_1)\, d_2/\sigma$	$\partial\Delta/\partial\sigma$
Volga	$\text{Vega} \cdot d_1 d_2/\sigma$	$\partial\text{Vega}/\partial\sigma$

Charm, speed, zomma, color and veta are also implemented in closed form.

2.2 Phoenix autocall by Monte Carlo

Spot paths follow geometric Brownian motion with drift $r - q$, simulated at the observation dates with antithetic variates. At each date t_j with performance $P_j = S_{t_j}/S_0$:

- **Early redemption:** if $P_j \geq$ the autocall barrier (and j is not the last date), the notional is repaid at t_j.
- **Coupon with memory:** if $P_j \geq$ the coupon barrier, the holder receives the coupon times the number of periods since the last payment. Missed coupons are paid when the barrier is next met.
- **At maturity:** if the product was never called, the notional is repaid when $P_n \geq$ the protection barrier, and $N \times P_n$ otherwise.

The price is the mean of discounted cash flows, $\sum_j c_j e^{-rt_j} + R\,e^{-r\tau}$, with a 95% interval $\pm 1.96\, s/\sqrt{n}$. Greeks are central differences of the price with the same seed in every bump (common random numbers), so the payoff discontinuities do not turn into noise.

2.3 Portfolio aggregation

For positions with quantity q_i, multiplier m_i and spot S:

$$\text{Delta-equivalent exposure} = \sum_i q_i m_i\, \Delta_i\, S,$$

$$\text{Gamma P\&L for } +1\% = \tfrac{1}{2} \sum_i q_i m_i\, \Gamma_i\, (0.01\, S)^2,$$

$$\text{Vega P\&L per vol point} = 0.01 \sum_i q_i m_i\, \mathcal{V}_i.$$

The stress grid uses, for each position, $\Delta\, dS + \frac{1}{2}\Gamma\, dS^2 + \mathcal{V}\, d\sigma + \text{Vanna}\; dS\, d\sigma$, with $dS = \text{shock} \times S$.

2.4 Delta-hedging simulation

A call is sold for its Black-Scholes premium. At each of n dates, the position holds the model delta in shares, pays a cost of c bp on the traded notional, and earns the dividend on the shares. At expiry, the replication error is the portfolio value minus the option payoff.

3 Results

Greeks. ATM call, 1 year, $r = 3\%$, $q = 2\%$, $\sigma = 22\%$: delta 0.5506, gamma 0.01756, vega 38.635, theta -0.0124 per day. ATM gamma at 1 month (0.08 year) is 0.06395 against 0.01203 at 2 years, a ratio of 5.32.

Phoenix autocall. Spot 100, volatility 25%, $r = 3\%$, $q = 2\%$, observations every six months for two years, 4% coupon per period, barriers 100% (call), 70% (coupon) and 60% (capital), 200,000 paths.

Quantity	Value
Price (% of notional)	100.71 (95% interval 100.64 to 100.77)
Called at 0.5 year	47.59%
Called at 1.0 year	11.93%
Called at 1.5 years	6.10%
Never called	34.38%
Capital loss, given never called	22.71%

Raising the call barrier from 95% to 105% moves the price from 100.51 to 101.05. Recomputed with the repository function (same seed), the price has delta 0.173 per unit of spot and vega −52.7 per 1.00 of volatility, so the holder is short volatility.

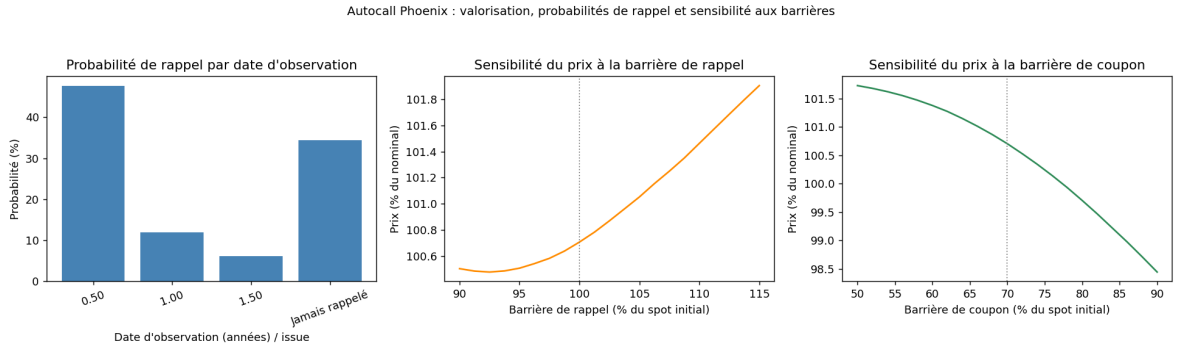


Figure 1: Phoenix autocall: probability of each outcome and price against the call and coupon barriers (labels in French).

Portfolio. Long 1,000 calls (strike 105) and short 600 puts (strike 95), both 1 year, plus 50 capital-protected notes and 30 capped bonus certificates, on one underlying (spot 100, volatility 22%). Delta-equivalent exposure is EUR 66,931. A +1% move in spot adds EUR 3.92 through gamma, and one volatility point adds EUR 235.88. The spot-by-volatility grid (spot ±20%, volatility ±10 points) goes from EUR −12,827 to +18,663.

Delta hedging. Short call, spot 100, strike 100, 6 months, volatility 25%, 40,000 paths. The error is in units of the option price:

Rehedges	Std, no cost	Mean, 10 bp cost	Std, 10 bp cost
1	5.3962	−0.1120	5.3962
4	2.8543	−0.1235	2.8641
12	1.6970	−0.1660	1.7075
52	0.8388	−0.2812	0.8523
252	0.3894	−0.5570	0.4407

More rehedging narrows the error but costs more. The mean cost grows from 0.11 to 0.56 between 1 and 252 rehedges.

4 Worked Example

Greeks by hand for the ATM call above ($S = K = 100$, $T = 1$, $r = 3\%$, $q = 2\%$, $\sigma = 22\%$):

$$\begin{aligned} d_1 &= \frac{(0.03 - 0.02 + 0.5 \times 0.22^2) \times 1}{0.22} = \frac{0.0342}{0.22} = 0.1555, \\ N(d_1) &= 0.5618, \quad \varphi(d_1) = 0.3942, \quad e^{-qT} = 0.9802, \\ \Delta &= 0.9802 \times 0.5618 = 0.5506, \\ \Gamma &= \frac{0.9802 \times 0.3942}{100 \times 0.22} = 0.01756, \\ \mathcal{V} &= 100 \times 0.9802 \times 0.3942 = 38.63. \end{aligned}$$

One Phoenix path (illustrative): performances at the four dates are 65%, 80%, 72%, 58%, with notional 100. The coupon is $4\% \times 100 = 4$ per period.

- $t = 0.5$: 65% is below the 70% coupon barrier, no coupon (one period is owed).
- $t = 1.0$: 80% is above 70%, so the coupon is paid for two periods: $4 \times 2 = 8$.
- $t = 1.5$: 72% is above 70%, coupon 4.
- $t = 2.0$: 58% is below 70%, no coupon. It is also below the 60% protection barrier, so redemption is $100 \times 0.58 = 58$.

Present value at $r = 3\%$: $8e^{-0.03} + 4e^{-0.045} + 58e^{-0.06} = 7.764 + 3.824 + 54.622 = 66.21$. The same path with a performance of 105% at the first date pays 4 and redeems 100 at $t = 0.5$.

5 Limits

- **Constant volatility:** no stochastic volatility or jumps. The volatility surface in the examples is a synthetic parametrisation, not market data.
- **No credit risk:** issuer risk is not in the structured product prices, although it is a large part of the risk of a note.
- **Barrier monitoring:** barriers are checked on the simulation grid, not continuously.
- **Portfolio example:** the bonus certificate is given a zero vega in the portfolio, so the EUR 235.88 per vol point leaves out its volatility sensitivity. The stress grid is a second-order expansion with a vanna term, not a full revaluation.
- **Hedging scaling:** the error does not fall exactly as $1/\sqrt{n}$. From 1 to 252 rehedges, the rule gives $5.3962/\sqrt{252} = 0.340$ and the simulation gives 0.389. These are model outputs, not recommendations.